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AI Genius

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Getting Started with Microsoft Agent Framework: **Build Practical AI Agents**

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Agenda

1. Introduction to Agents
2. Microsoft Agent Framework Overview
3. Demo: Building Your First Agent
4. Demo: Travel Assistant
5. Q&A

Anatomy of AI Agent

Language Model

Core reasoning and decision-making engine

Safety & Control

Human-in-the-loop workflows, guardrails, constraints

Security & Compliance

Data protection, access control, audit trails

Context Management

Memory systems, knowledge base, state management

Cost Optimisation

Efficient tool usage, model selection, caching strategies

Tools

Functions, MCP tools, external APIs, code interpreter

Observability

Monitoring agent behavior and performance

Prompt Engineering

Template design, few-shot examples, dynamic prompting

Collaboration

Multi-agent communication, A2A protocols, workflow orchestration, shared state

Testing & Evaluation

Measure against benchmarks, continuous improvement

Agent Loop - Understanding How Agents Run

Agent Loop

Repeats reasoning → tool use → memory update until a final response is produced

Language Model

Interprets user input, reasons, and generates responses

Tools

Executes actions and retrieves external data

Memory

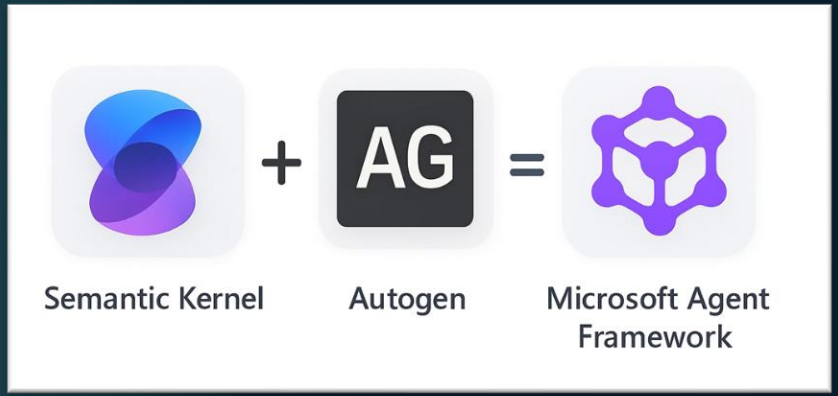
Maintains context across interactions

When is my flight to Sydney?

User is asking about their personal travel itinerary.
This requires accessing user-specific booking data.
I need to identify the user and query their bookings.
Call get_user_identity
Call get_bookings(user_id, type: "flight", destination: "Sydney")
Execute tool call → Get User Identity → Fetch booking details

Your flight to Sydney is on March 15th at 2:30 PM with Qantas (QF401).

Microsoft Agent Framework



Unified Platform
One SDK combining Semantic Kernel and AutoGen for both single and multi-agent workflows.

Multi-Language Support
Native support for .NET and Python

Extensible Architecture
Swap memory providers, LLMs, and tools without rewriting your entire application.

Open Standards
Supports industry-standard protocols MCP for tool integration, A2A for agent communication, and AG-UI for frontend interaction.

Observability
Trace conversations, tool calls, and agent handoffs using OpenTelemetry

Hosting Options
Agents run as OpenAI-compatible HTTP endpoints, no custom API development needed

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Demo: Building Your First Agent

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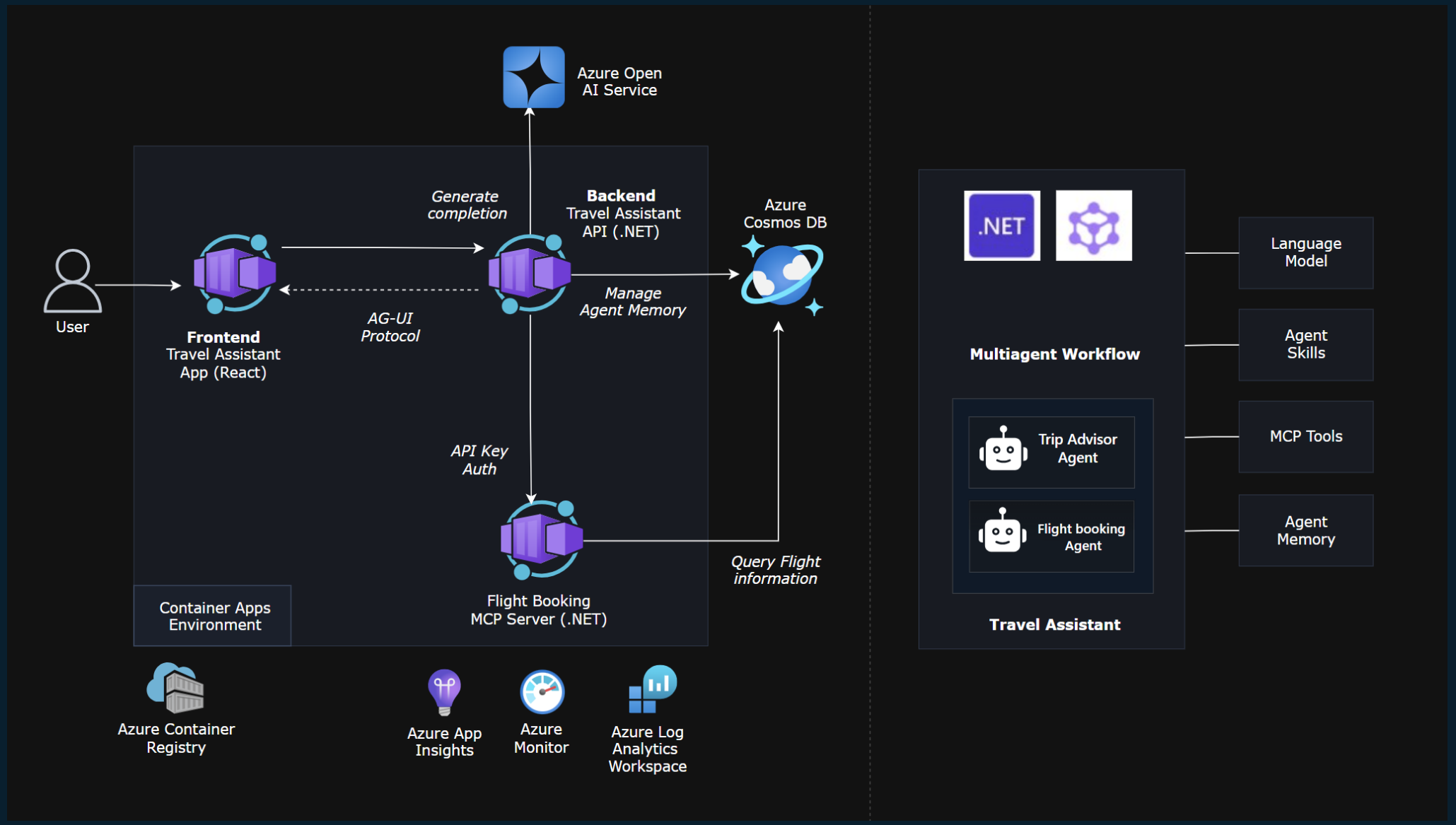
Demo:
Travel Assistant

Travel Assistant Architecture



How Agents Remember

Short-Term Memory

- Remembers information within a single session
- Maintains conversation context
- Enables coherent responses

Long-Term Memory

- Persists information across sessions
- Stores user preferences
- Supports personalization over time

Episodic memory

- Enables context-aware responses based on past events
- Captures specific events or interactions

Why Agentic Apps Need AG-UI

Standardization
Open protocol.
Build once, use with any AG-UI compatible frontend

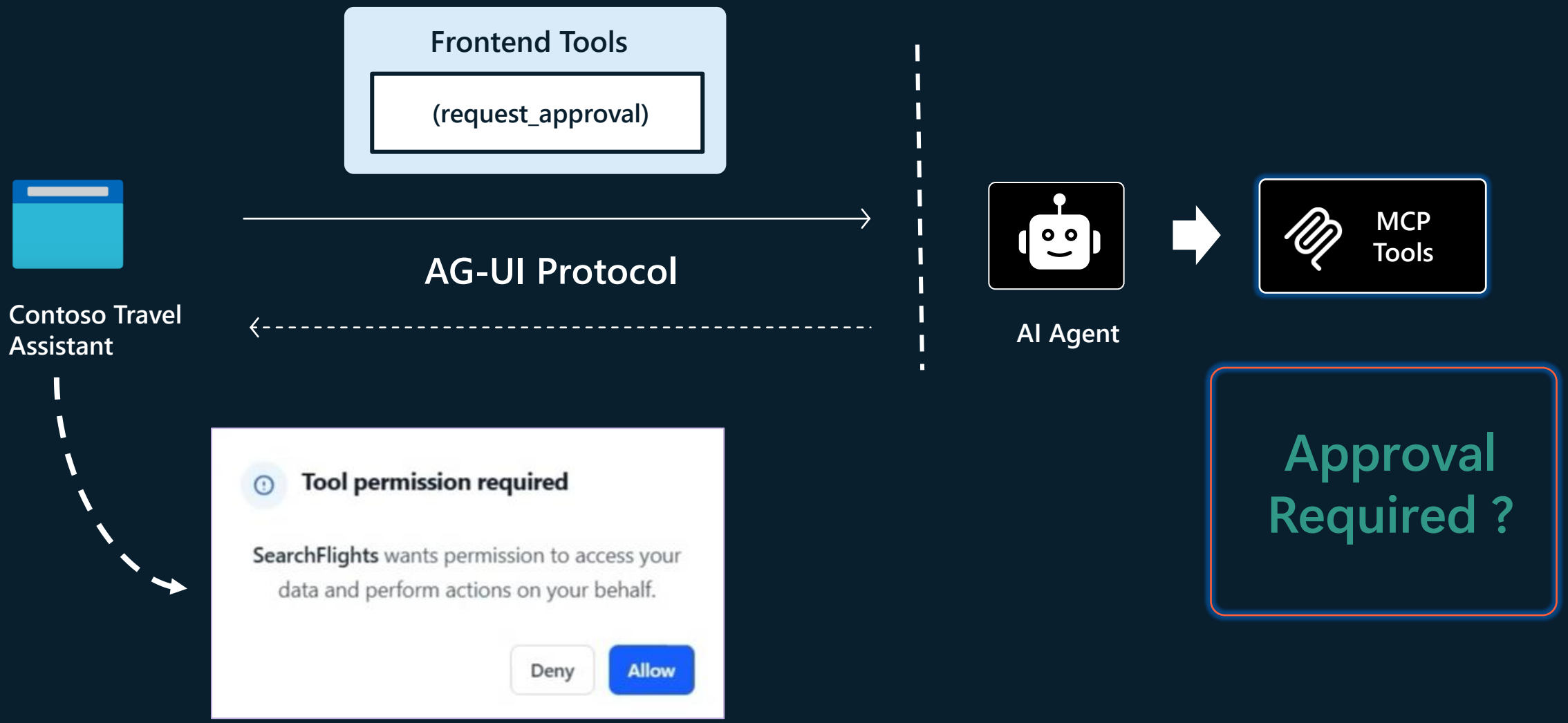
Stateful Conversations
Maintains session state between client and server

Real-Time Updates
Stream agent thoughts and intermediate steps

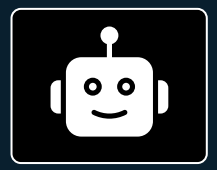
Rich, Dynamic UI Components
Support for images, videos, custom components

Human-in-the-Loop Workflows
Built-in support for approval requests

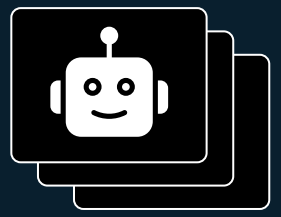
Human In The Loop Approval (HITL) Workflow



Human In The Loop Approval (HITL) Workflow



Triage Agent



Trip Advisor Agent

Flight Booking Agent

Agent N

Triage Agent

Triage the user query and direct to the right specialized agent

Multiple Specialized Agents

Each agent designed to perform specific tasks efficiently

Handoff Workflow Pattern

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Resources and Further Learning

Get the **repo** for this session:

binarytrails-ai/aigenius-maf-travel-assistant

Explore more samples -

github.com/microsoft/agent-framework

Read the Agent Framework SDK Docs:

aka.ms/agent-framework

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Thank You!